

TE Curve Verification Pipeline Selected Active Model Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report evaluates only the currently selected active model families against the repository held-out TE-curve test split. The `global` surface is intentionally excluded from this reduced decision report.

Dataset And Split

- dataset config: `config/datasets/transmission_error_dataset.yaml` ;
- dataset root: `data\polished_dataset` ;
- comparison mode: `selected_active_model_matrix` ;
- candidate count: `7` ;
- held-out curve count before candidate filtering: `94` ;
- percentage-error denominator: `peak_to_peak_truth` ;
- evaluated surface: `Bw` ;
- evaluated direction: `backward` ;

Candidate Inventory

Candidate	Source	Family	Surface	Valid Directions
polished_setpoints_feedforward_Bw	polished_setpoints_model_archive	feedforward	Bw	backward
polished_setpoints_tree_Bw	polished_setpoints_model_archive	tree	Bw	backward
polished_setpoints_harmonic_regression_Bw	polished_setpoints_model_archive	harmonic_regression	Bw	backward
polished_setpoints_periodic_mlp_harmonic_Bw	polished_setpoints_model_archive	periodic_mlp_harmonic	Bw	backward
polished_setpoints_periodic_gru_sequence_Bw	polished_setpoints_model_archive	periodic_gru_sequence	Bw	backward
polished_setpoints_wave4_1_mae_robust_loss_Bw	polished_setpoints_model_archive	wave4_1_mae_robust_loss	Bw	backward
polished_setpoints_wave4_2_quantile_p10_p50_p90_Bw	polished_setpoints_model_archive	wave4_2_quantile_p10_p50_p90	Bw	backward

Exact Model Paths

Candidate	Family	Surface	ONNX Model Path	Python Model Path
polished_setpoints_feedforward_Bw	feedforward	Bw	models/polished_dataset/setpoints/exported/feedforward/backward/2026-07-07-17-41-33_te_feedforward_bw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/feedforward/backward/2026-07-07-17-41-33_te_feedforward_bw_polished_setpoints/python/feedforward-epoch=057-val_mae=0.00164066.ckpt
polished_setpoints_tree_Bw	tree	Bw	models/polished_dataset/setpoints/exported/tree/backward/2026-07-07-09-37-16_te_tree_bw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/tree/backward/2026-07-07-09-37-16_te_tree_bw_polished_setpoints/python/tree_model.pkl
polished_setpoints_harmonic_regression_Bw	harmonic_regression	Bw	models/polished_dataset/setpoints/exported/harmonic_regression/backward/2026-07-07-23-39-40_te_harmonic_regression_bw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/harmonic_regression/backward/2026-07-07-23-39-40_te_harmonic_regression_bw_polished_setpoints/python/harmonic_regression-epoch=044-val_mae=0.01715066.ckpt
polished_setpoints_periodic_mlp_harmonic_Bw	periodic_mlp_harmonic	Bw	models/polished_dataset/setpoints/exported/periodic_mlp_harmonic/backward/2026-07-08-02-29-32_te_periodic_mlp_harmonic_bw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/periodic_mlp_harmonic/backward/2026-07-08-02-29-32_te_periodic_mlp_harmonic_bw_polished_setpoints/python/periodic_mlp-epoch=081-val_mae=0.00121896.ckpt
polished_setpoints_periodic_gru_sequence_Bw	periodic_gru_sequence	Bw	models/polished_dataset/setpoints/exported/periodic_gru_sequence/backward/2026-07-08-23-18-30_te_periodic_gru_sequence_bw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/periodic_gru_sequence/backward/2026-07-08-23-18-30_te_periodic_gru_sequence_bw_polished_setpoints/python/periodic_gru_sequence-epoch=064-val_mae=0.00189811.ckpt
polished_setpoints_wave4_1_mae_robust_loss_Bw	wave4_1_mae_robust_loss	Bw	models/polished_dataset/setpoints/exported/wave4_1_mae_robust_loss/backward/2026-07-12-16-51-03_te_wave4_1_mae_robust_loss_bw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave4_1_mae_robust_loss/backward/2026-07-12-16-51-03_te_wave4_1_mae_robust_loss_bw_polished_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=088-val_mae=0.00183184.ckpt
polished_setpoints_wave4_2_quantile_p10_p50_p90_Bw	wave4_2_quantile_p10_p50_p90	Bw	models/polished_dataset/setpoints/exported/wave4_2_quantile_p10_p50_p90/backward/2026-07-14-13-54-54_te_wave4_2_quantile_p10_p50_p90_bw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave4_2_quantile_p10_p50_p90/backward/2026-07-14-13-54-54_te_wave4_2_quantile_p10_p50_p90_bw_polished_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=063-val_mae=0.00181729.ckpt

Metric Ranking

Rank	Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
1	polished_setpoints_tree_Bw	0.003928	0.004502	7.417	14.932
2	polished_setpoints_wave4_1_mae_robust_loss_Bw	0.006179	0.006697	12.394	23.419
3	polished_setpoints_periodic_gru_sequence_Bw	0.007298	0.007811	14.582	29.283
4	polished_setpoints_wave4_2_quantile_p10_p50_p90_Bw	0.008246	0.008717	16.914	29.922
5	polished_setpoints_periodic_mlp_harmonic_Bw	0.009651	0.010153	19.771	37.058
6	polished_setpoints_feedforward_Bw	0.010159	0.010736	20.741	42.533
7	polished_setpoints_harmonic_regression_Bw	0.017296	0.017715	37.398	77.577

Direction Breakdown

Direction	Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
backward	polished_setpoints_tree_Bw	0.003928	0.004502	7.417	14.932
backward	polished_setpoints_wave4_1_mae_robust_loss_Bw	0.006179	0.006697	12.394	23.419
backward	polished_setpoints_periodic_gru_sequence_Bw	0.007298	0.007811	14.582	29.283
backward	polished_setpoints_wave4_2_quantile_p10_p50_p90_Bw	0.008246	0.008717	16.914	29.922
backward	polished_setpoints_periodic_mlp_harmonic_Bw	0.009651	0.010153	19.771	37.058
backward	polished_setpoints_feedforward_Bw	0.010159	0.010736	20.741	42.533
backward	polished_setpoints_harmonic_regression_Bw	0.017296	0.017715	37.398	77.577

Curve Evidence

Each selected candidate is shown with the same four deterministic held-out operating conditions for this direction. The dark line is the measured TE curve and the blue line is the model prediction for the same operating condition.

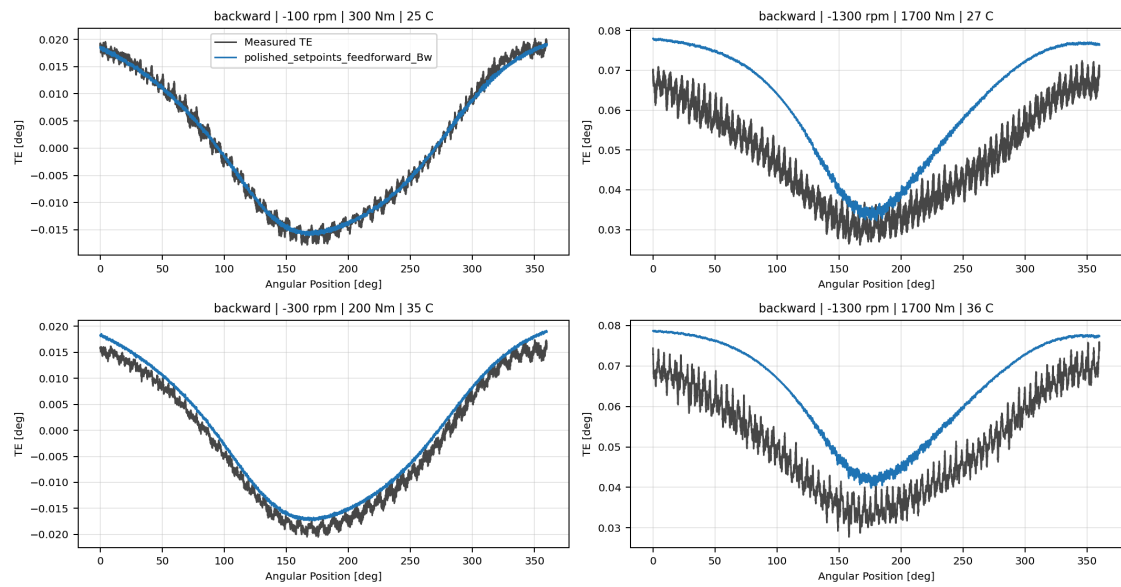
Shared operating conditions

- 100 rpm, 300 Nm, 25 C
- 1300 rpm, 1700 Nm, 25 C
- 300 rpm, 200 Nm, 35 C
- 1300 rpm, 1700 Nm, 35 C

Candidate	Role	Curve MAE [deg]	Curve RMSE [deg]	Mean MPE [%]	P95 MPE [%]
polished_setpoints_feedforward_Bw	Selected candidate	0.010159	0.010736	20.741	42.533
polished_setpoints_tree_Bw	Surface leader	0.003928	0.004502	7.417	14.932
polished_setpoints_harmonic_regression_Bw	Selected candidate	0.017296	0.017715	37.398	77.577
polished_setpoints_periodic_mlp_harmonic_Bw	Selected candidate	0.009651	0.010153	19.771	37.058
polished_setpoints_periodic_gru_sequence_Bw	Selected candidate	0.007298	0.007811	14.582	29.283
polished_setpoints_wave4_1_mae_robust_loss_Bw	Selected candidate	0.006179	0.006697	12.394	23.419
polished_setpoints_wave4_2_quantile_p10_p50_p90_Bw	Selected candidate	0.008246	0.008717	16.914	29.922

polished_setpoints_feedforward_Bw

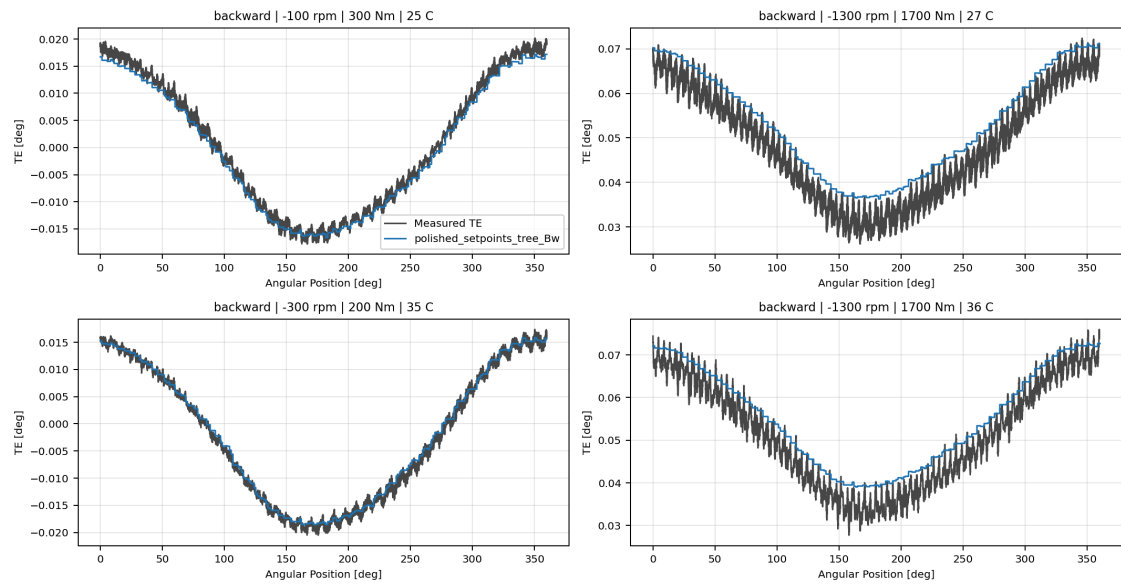
polished_setpoints_feedforward_Bw



polished_setpoints_feedforward_Bw measured-versus-predicted TE collage

polished_setpoints_tree_Bw

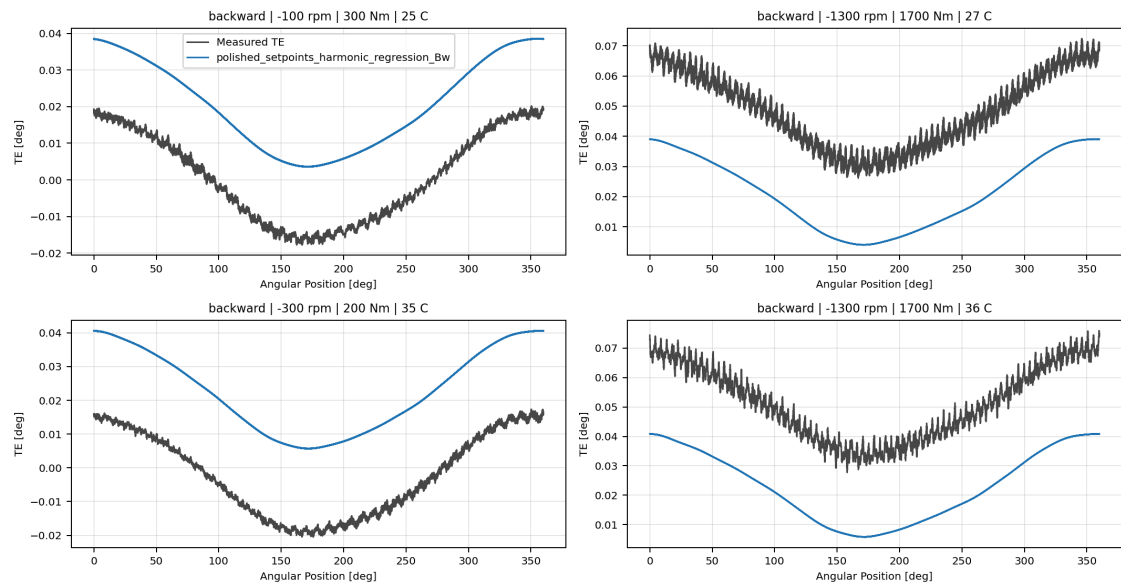
polished_setpoints_tree_Bw



polished_setpoints_tree_Bw measured-versus-predicted TE collage

polished_setpoints_harmonic_regression_Bw

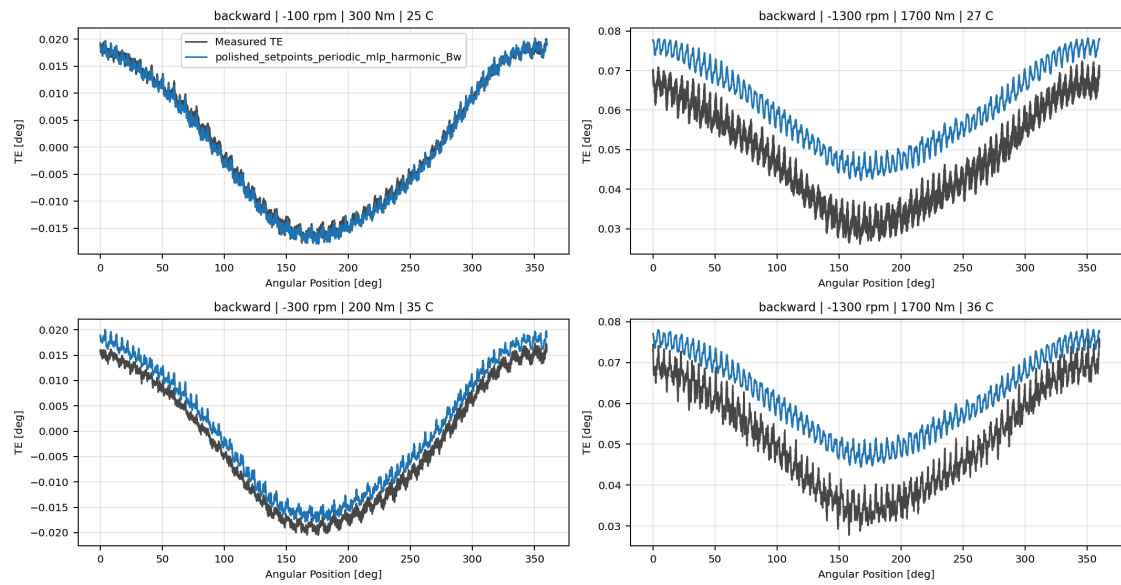
polished_setpoints_harmonic_regression_Bw



polished_setpoints_harmonic_regression_Bw measured-versus-predicted TE collage

polished_setpoints_periodic_mlp_harmonic_Bw

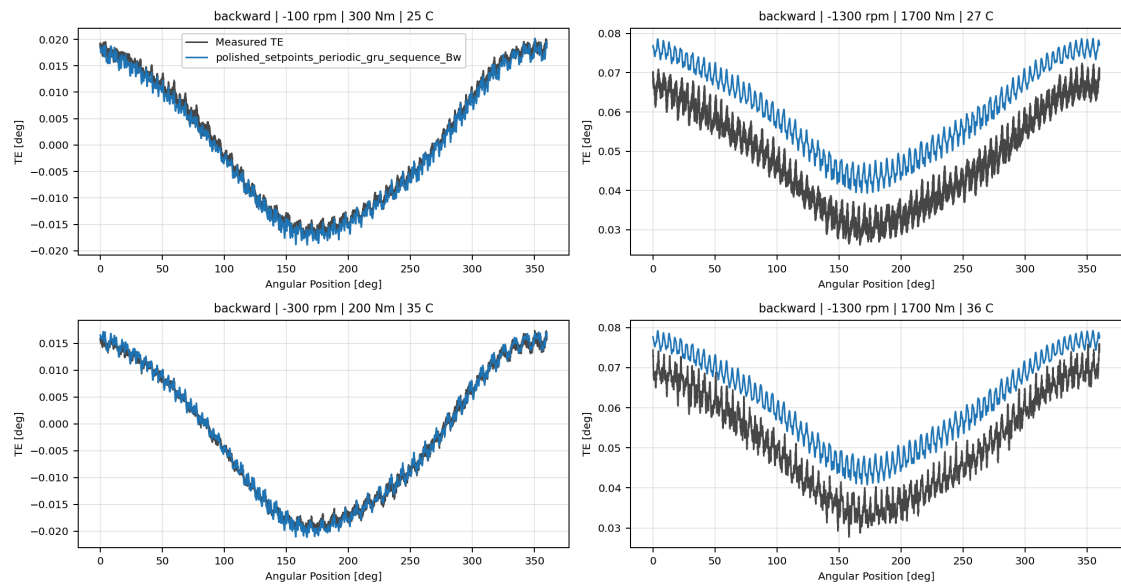
polished_setpoints_periodic_mlp_harmonic_Bw



polished_setpoints_periodic_mlp_harmonic_Bw measured-versus-predicted TE collage

polished_setpoints_periodic_gru_sequence_Bw

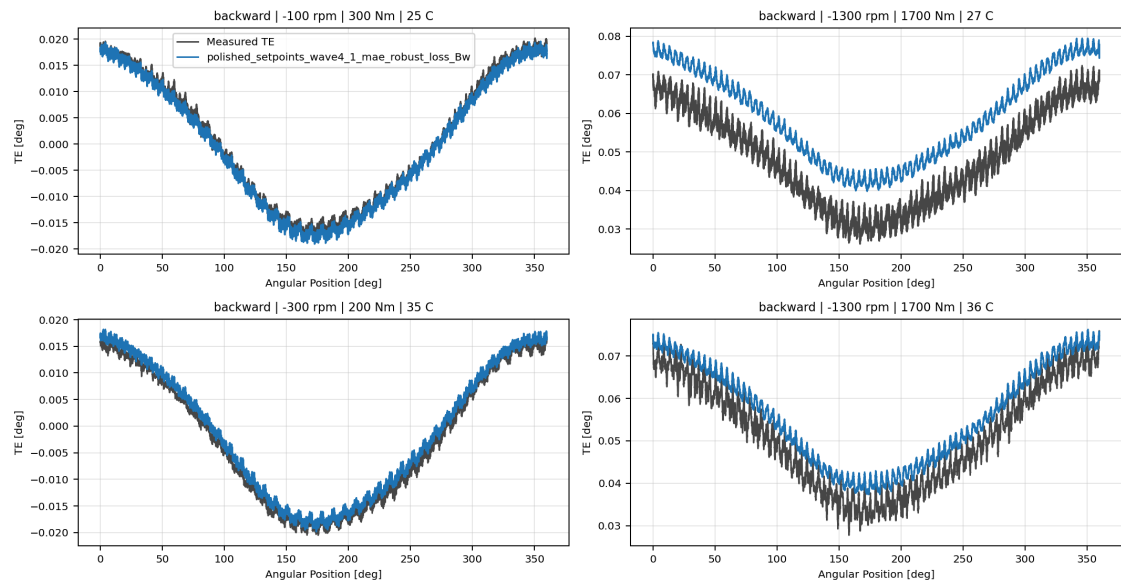
polished_setpoints_periodic_gru_sequence_Bw



polished_setpoints_periodic_gru_sequence_Bw measured-versus-predicted TE collage

polished_setpoints_wave4_1_mae_robust_loss_Bw

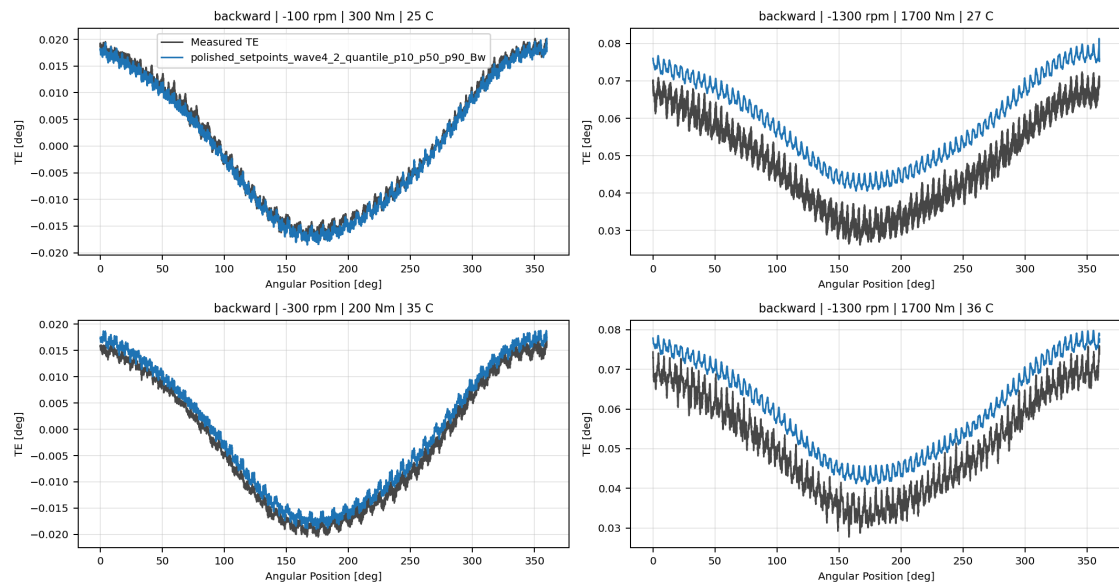
polished_setpoints_wave4_1_mae_robust_loss_Bw



polished_setpoints_wave4_1_mae_robust_loss_Bw measured-versus-predicted TE collage

polished_setpoints_wave4_2_quantile_p10_p50_p90_Bw

polished_setpoints_wave4_2_quantile_p10_p50_p90_Bw



polished_setpoints_wave4_2_quantile_p10_p50_p90_Bw measured-versus-predicted TE collage

Artifacts

- summary YAML: `output\validation_checks\track2_reference_comparison\2026-07-19-17-44-49_track2_selected_active_polished_setpoints_matrix_track2_selected_active_polished_setpoints_backward_2026_07_19\validation_summary.yaml` ;
- per-condition CSV: `output\validation_checks\track2_reference_comparison\2026-07-19-17-44-49_track2_selected_active_polished_setpoints_matrix_track2_selected_active_polished_setpoints_backward_2026_07_19\per_condition_metrics.csv` ;
- grouped report plot root: `doc\reports\campaign_results\track_2\verification_plots\selected_active_model_matrix` ;
- grouped report plot count: `0` ;

Interpretation

Rows are ranked by mean percentage error within this selected-active surface report. The curve-evidence section uses shared deterministic held-out operating conditions so visual shape fidelity can be compared across the selected families.

Open Gaps

- This remains an offline TE-curve comparison and does not replace the future online [Table 9](#) compensation benchmark.
- RCIM Model-Bank Reproduction remains a separate paper-reference benchmark path and is not part of this selected-active model-only report.